Exp 5: ULTRASONIC SENSOR

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1. Introduction:

This project demonstrates how to measure distance using an ultrasonic sensor with the Raspberry Pi Pico. An ultrasonic sensor works by transmitting high-frequency sound waves and measuring the time taken for the echo to return after reflecting from an object.

The sensor uses two main pins: Trig (Trigger) to send the ultrasonic pulse and Echo to receive the reflected signal. The measured time is used to calculate the distance between the sensor and the object.

2. Scope:

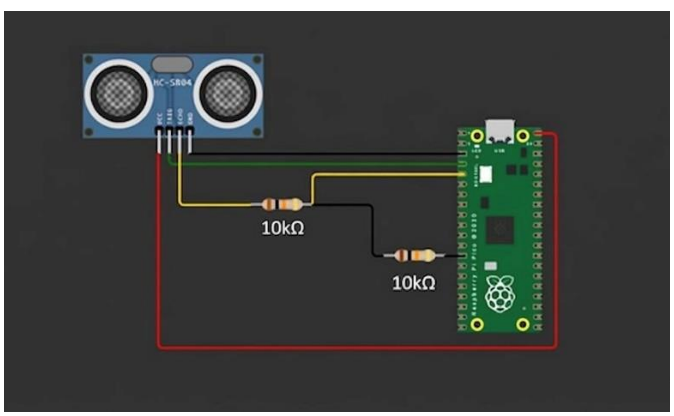
This project provides an understanding of distance measurement using ultrasonic sensors and real-time data monitoring. It introduces concepts such as time-based measurement, sensor interfacing, and serial data output. These concepts are widely used in robotics, obstacle detection systems, and automation applications.

3. Prerequisite:

Hardware: Raspberry Pi Pico Microcontroller, USB Connector, Breadboard, Ultrasonic Sensor (HC-SR04), Jumper Wires

Software: VS Code (with Pico SDK), PuTTY Serial Console (in Windows)

4. Connection Diagram:



5. Working Principle:

The ultrasonic sensor is connected to the Raspberry Pi Pico using two GPIO pins: Trig and Echo. To start measurement, a short pulse of approximately 10 microseconds is sent to the Trig pin. This triggers the sensor to emit an ultrasonic sound wave.

The sound wave travels through the air, reflects off an object, and returns to the sensor. The Echo pin goes HIGH for the duration of the travel time of the sound wave. The microcontroller measures this time interval.

Using the measured time and the speed of sound (approximately 343 m/s), the distance is calculated using the formula:

Distance = (Time × Speed of Sound) / 2

The division by 2 accounts for the round-trip travel of the sound wave. The calculated distance is then continuously printed on the UART serial console. As the object moves closer or farther, the distance value changes accordingly in real time.

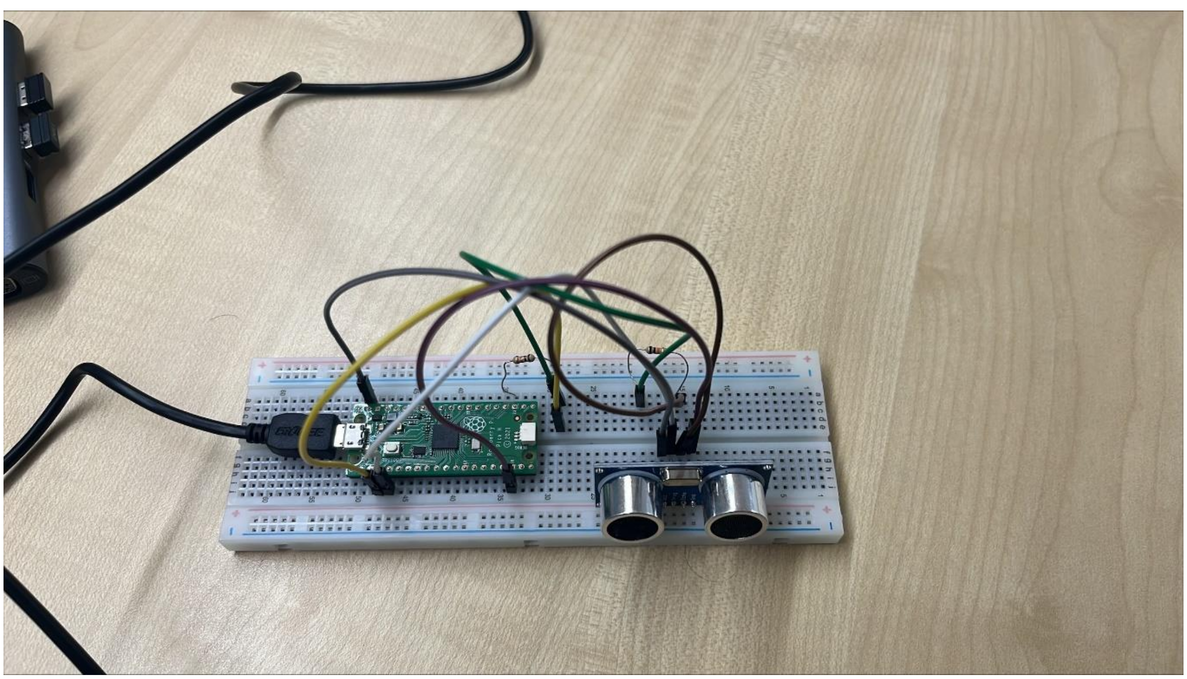
6. Application code:

```c
#include <stdio.h>
#include "pico/stdlib.h"

#define TRIG_PIN 3
#define ECHO_PIN 2

float get_distance() {
    gpio_put(TRIG_PIN, 0);
    sleep_us(2);
    gpio_put(TRIG_PIN, 1);
    sleep_us(10);
    gpio_put(TRIG_PIN, 0);
    while (gpio_get(ECHO_PIN) == 0);
    absolute_time_t start = get_absolute_time();
    while (gpio_get(ECHO_PIN) == 1);
    absolute_time_t end = get_absolute_time();
    int64_t duration = absolute_time_diff_us(start, end);
    float distance = (duration * 0.0343) / 2;
    return distance;
}

int main() {
    stdio_init_all();
    gpio_init(TRIG_PIN);
    gpio_set_dir(TRIG_PIN, GPIO_OUT);
    gpio_init(ECHO_PIN);
    gpio_set_dir(ECHO_PIN, GPIO_IN);
    while (1) {
        float dist = get_distance();
        printf("Distance: %.2f cm\n", dist);
        sleep_ms(500);
    }
}
```

7. Output:

```
Distance: 38.55 cm
Distance: 56.49 cm
Distance: 41.93 cm
Distance: 14.42 cm
Distance: 4.37 cm
Distance: 1.84 cm
Distance: 2.02 cm
Distance: 3.53 cm
Distance: 2.50 cm
Distance: 5.87 cm
Distance: 10.94 cm
Distance: 13.31 cm
Distance: 17.05 cm
Distance: 23.43 cm
Distance: 22.48 cm
Distance: 23.51 cm
Distance: 25.42 cm
Distance: 29.74 cm
Distance: 31.04 cm
Distance: 33.49 cm
Distance: 33.00 cm
Distance: 29.93 cm
Distance: 63.52 cm
Distance: 57.44 cm
```

8. Hardware Image:



9. Appendix:

References

- Official Documentation for Raspberry Pi Pico and Pico W
- Technical Datasheet for Raspberry Pi Pico
- GitHub Repository for Raspberry Pi Pico SDK
- Ultrasonic Sensor (HC-SR04) Datasheet
- PuTTY Serial Console Documentation